

So, as consumers download games that can be played offline, access the stock market, and pay for groceries with their wireless devices, business people will read e-mail, send short messages, and read graphics and charts on their wireless devices. Unfortunately, this wireless utopia is unlikely to come without a price—increasingly sophisticated wireless threats that utilise the same capabilities (e.g., connectivity, functionality, and speed). Viruses can spread from wireless device to wireless device, from wireless device to point-of-sale device (e.g., at the grocery counter), and from wireless device to PC.

The latter path offers a mode of transmission for viruses to wireless and wired internal LANs, and further propagation across the Internet. Currently, corporate IT managers have little control over which wireless and handheld devices their users are connecting to the network. Connecting a portable device (such as a PDA) into a PC that is connected (or subsequently connected) to the network is similar to inserting a floppy disk—that has not been scanned for viruses—into a computer.

A protection solution for the wireless infrastructure must have the following attributes:

- Multiple layers of protection to address the various entry points and transmission paths of viruses and malicious code
- Integration of centralised management of all antivirus solutions including maintenance of gateway, server, desktop, and device-level protection
- Implementation within the wireless infrastructure for early detection to minimise damage and costs
- Tools tailored to the wireless threat, rather than merely applying wired world tools
- Mechanisms for automatic maintenance, updating, and upgrading of virus protection since such protection is only as good as the last update
- Involve all parties via increased awareness of the potential threat including corporate IT managers, service providers, operating system and application developers, and end users